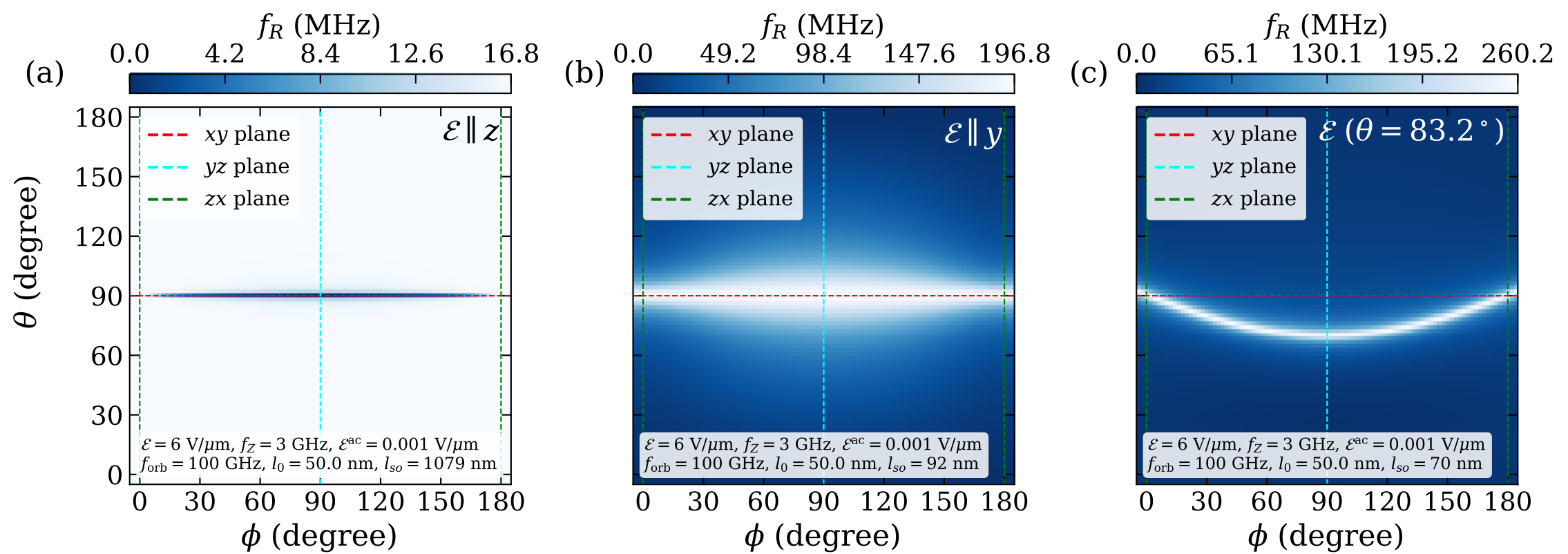
(a)
f_R (MHz)
0.0 4.2 8.4 12.6 16.8
θ (degree)
180 150 120 90 60 30 0
xy plane
yz plane
zx plane
$\mathcal{E} \parallel z$
$\mathcal{E} = 6$ V/μm, $f_Z = 3$ GHz, $\mathcal{E}^{\rm ac} = 0.001$ V/μm
$f_{\rm orb} = 100$ GHz, $l_0 = 50.0$ nm, $l_{so} = 1079$ nm
0 30 60 90 120 150 180
ϕ (degree)
(b)
f_R (MHz)
0.0 49.2 98.4 147.6 196.8
xy plane
yz plane
zx plane
$\mathcal{E} \parallel y$
$\mathcal{E} = 6$ V/μm, $f_Z = 3$ GHz, $\mathcal{E}^{\rm ac} = 0.001$ V/μm
$f_{\rm orb} = 100$ GHz, $l_0 = 50.0$ nm, $l_{so} = 92$ nm
0 30 60 90 120 150 180
ϕ (degree)
(c)
f_R (MHz)
0.0 65.1 130.1 195.2 260.2
xy plane
yz plane
zx plane
$\mathcal{E}$ ($\theta = 83.2^\circ$)
$\mathcal{E} = 6$ V/μm, $f_Z = 3$ GHz, $\mathcal{E}^{\rm ac} = 0.001$ V/μm
$f_{\rm orb} = 100$ GHz, $l_0 = 50.0$ nm, $l_{so} = 70$ nm
0 30 60 90 120 150 180
ϕ (degree)